CampusCare

Automated University Facility Issue Reporting, Tracking & SLA Management System

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Presentation Agenda

Structured Overview of CampusCare Project

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- Followed by Live System Demonstration

1. Introduction

Centralized Intelligent Facility Management for Campuses

What is CampusCare?

- A unified **3-tier web platform** designed for university campuses to streamline facility maintenance and issue tracking.
- Eliminates chaotic paper logbooks, unmonitored WhatsApp messages, and delayed verbal complaints.
- Connects **Students**, **Department Staff Resolvers**, and **Administrative Overseers** in a single automated lifecycle.

Core System Capabilities

- **Anonymous & Authenticated Reporting:** Students submit complaints with image evidence and exact campus room locations.
- **Automated SLA Resolution Windows:** Priority-based countdown timers (4h Critical, 24h High, 72h Medium).
- **AI Operations Dispatcher:** Live, context-aware ticket assistant with instant human takeover.
- **Transparent Audit Trails:** Tamper-proof history tracking every status change and assignment.

2. Problem Statement

Operational Bottlenecks in Traditional Campus Maintenance

Fragmented Channels

Complaints are submitted randomly across phone calls, unofficial Facebook groups, personal messages, and front-desk physical paper books.

Zero Status Visibility

Students have no mechanism to track if their issue was acknowledged, who is assigned, or when repairs will be finished.

Lack of Accountability

Staff departments have no centralized task queue. Unresolved tickets get lost between departments with no SLA enforcement.

No Analytical Data

University management lacks historical metrics to identify chronic equipment failures or measure department repair efficiency.

3. Project Objectives

Target Goals and Measurable Deliverables

Centralized Portal

Build a role-based single-entry portal where students log issues across departments (IT, Electrical, HVAC, Sanitation, Lost & Found).

Strict SLA Control

Implement an automated Service Level Agreement engine calculating real-time resolution deadlines with visual overdue indicators.

AI & Live Chat

Deploy an AI Dispatcher providing instant contextual ticket updates, with instant human staff takeover capabilities.

Measurable Reliability Benchmarks

Zero-Data-Loss Transactions: Guaranteed database state consistency with PostgreSQL RLS.
Sub-Second Latency: In-memory and Upstash Redis caching for real-time ticket messaging.
100% Core Verification: Rigorous unit test coverage validating Zod schemas and RBAC security rules.

4. Related Works & Gap Analysis

Benchmarking CampusCare Against Existing Solutions

Platform	Primary Strength	Identified Gap for University Campuses
Jira Service Management	Enterprise ticketing & ITIL workflows	Cost-prohibitive for universities; lacks campus location hierarchy & student anonymity.
Freshservice / Zendesk	Automated email workflows & CRM	Complex configuration; no integrated campus room/lab tagging or student feedback loops.
UITS Facebook Groups	High student engagement (8k+ members)	Unstructured text; complaints buried in feeds; zero accountability or resolution tracking.
CampusCare (Proposed)	**Campus-native role architecture, SLA engine, AI live dispatcher**	**100% Open-Source, zero licensing cost, built specifically for academic infrastructure.**

5. Development Methodology

Agile SDLC with Continuous Automated Verification

Agile Iterative Sprints

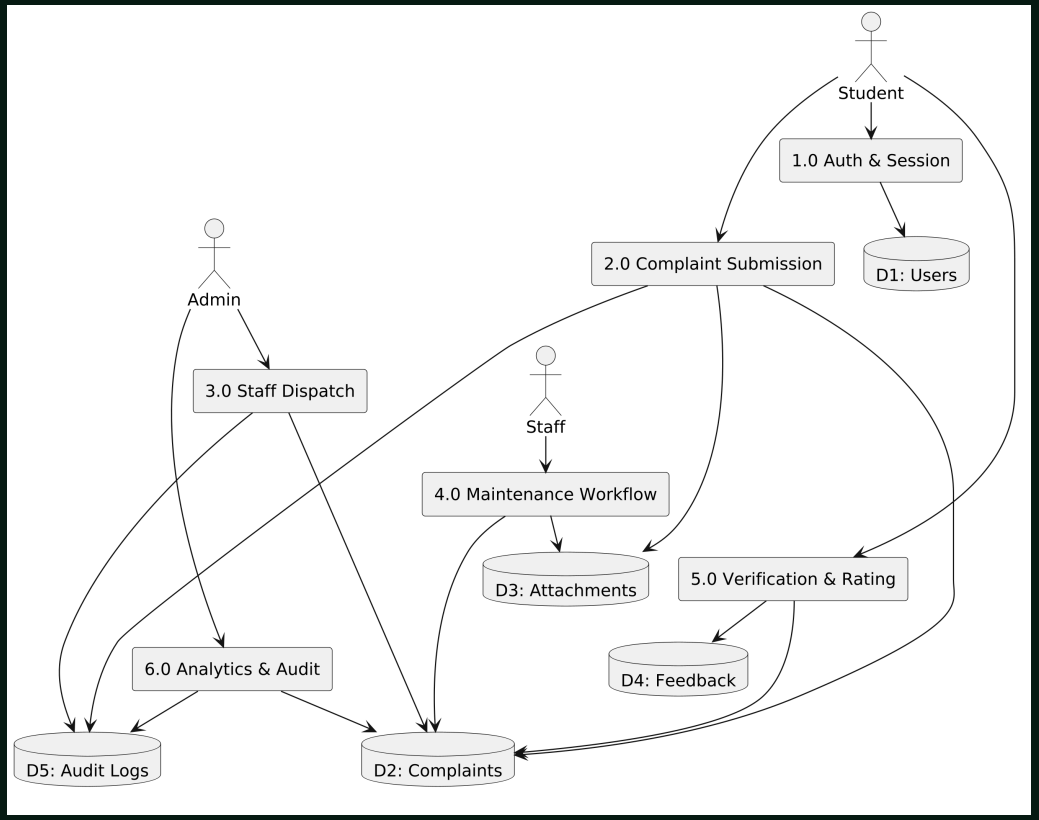
- **Sprint 1:** Requirements engineering & SRS documentation.
- **Sprint 2:** Supabase DB schema, migrations & RLS policies.
- **Sprint 3:** Next.js App Router UI, forms & validation.
- **Sprint 4:** SLA deadline engine & staff resolution workflows.
- **Sprint 5:** Redis live chat, Gemini AI Dispatcher & QA.

QA & Verification Loop

- **Formal Unit Testing:** Vitest suite with 19 automated test cases executed in under 10ms.
- **100% Code Coverage:** 100% Stmts, Branches, and Functions verified across core business logic (`rbac.ts`, `sla.ts`, `validations/`).
- **Snyk Vulnerability Scan:** Scanned all dependencies; verified 0 high-severity security vulnerabilities.

6. 3-Tier System Architecture

Clean Separation of Presentation, Business Logic & Data Persistence



Presentation Tier

Next.js 16 App Router, Tailwind CSS, Turbopack, responsive mobile-first UI.

Application / Logic Tier

Next.js Server Actions, Zod Schema Validator, SLA Engine, Gemini 3.5 Dispatcher.

Data Persistence Tier

PostgreSQL with Row Level Security, Upstash Redis pub/sub live chat, Supabase Storage.

7. Technology Stack & Tools

Modern Cloud-Native & Open-Source Toolchain

Frontend & Full-Stack

- **Next.js 16 (App Router):** Server components and optimized Turbopack build engine.
- **TypeScript & React 19:** Strict static typing preventing runtime null-pointer exceptions.
- **Tailwind CSS:** Custom glassmorphism design system tailored for academic readability.

Database & Security

- **Supabase PostgreSQL:** Managed database with Row-Level Security (RLS) enforcement.
- **Upstash Redis:** Low-latency memory store powering live chat with global TTL expiration.
- **Supabase Storage:** Secure cloud bucket for complaint image evidence attachments.

Artificial Intelligence & QA

- **Google Gemini AI:** High-throughput multi-model cascade (`gemini-3.5-flash-lite`).
- **Vitest & V8 Coverage:** Ultra-fast unit testing framework (19 test cases, 100% coverage).
- **Snyk & Qodo Gen:** Static security scanning and boundary test case generation.

Documentation & Modeling

- **Typst 0.15:** Publication-grade document and presentation compilation engine.
- **PlantUML & Mermaid:** Formal UML modeling (Class, ER, DFD, Sequence, State).

8. Core Feature: Context-Grounded AI Dispatcher

Intelligent Ticket Assistance with Zero Hallucinations & Human Handover

How the AI Dispatcher Covers Ongoing Tickets

- **Real-Time Ticket Grounding:** Unlike generic chatbots, the AI ingests live database fields for each ticket:
  - Current Status (`Submitted`, `In Progress`, `Resolved`)
  - Exact Location (`Lab Room 402, Building B`)
  - SLA Target Deadline countdown
  - Recent staff progress notes & activity feed
- **Continuous Student Inquiry Answering:** When a student asks **“When will this be fixed?”** or **“Who has my lost ID?”**, the AI reads actual staff logs and responds accurately without guessing.
- **Multi-Turn Conversational Memory:** Remembers previous questions in the live chat session for natural dialogue.

High-Availability Architecture

- **4-Second Multi-Model Cascade:** Prioritizes low-latency models (`gemini-3.5-flash-lite`, 1.1s) with automatic failover to `gemini-3.8-flash`.
- **Instant Human Handover:**
  - Clicking **Talk to Staff** instantly silences the AI bot.
  - Staff can click **Join Live Chat** from `/staff/dashboard` to speak directly with the student.
- **Resilient Fallback:** If cloud endpoints disconnect, a smart local fallback answers based on ticket context.

9. Outcome & Impact Analysis

Measurable Value Across Triple Bottom Line

Economic Impact

- **Zero Licensing Fees:** 100% open-source software stack saves university IT budget.
- **Preventative Cost Savings:** Fast reporting of water leaks and electrical faults prevents expensive building damages.
- **Optimized Labor:** Eliminates manual ticket sorting and phone follow-ups.

Environmental Impact

- **Paperless Operations:** Replaces paper requisition forms and physical front-desk logbooks.
- **Rapid Energy Conservation:** Quick repair of malfunctioning HVAC units and outdoor lights minimizes wasted campus electricity.

Social Sustainability

- **Student Empowerment:** Transparent timeline tracking restores student trust in campus administration.
- **Safe Anonymous Reporting:** Allows students to report sensitive issues (harassment, hazards) without fear.

10. Result Analysis & Performance

Empirical System Metrics and Validation Outcomes

Test Suite Execution (Vitest v4.1)

Module Tested	Tests	Status
Zod Input Validation	8	Passed (1ms)
SLA Deadline Engine	6	Passed (0ms)
RBAC & Admin Guard	5	Passed (0ms)
Total Passing Tests	**19 / 19**	**100% GREEN**

Performance Benchmarks

- **Production Build Time:** Compiled 12 routes in 30.8s via Next.js Turbopack.
- **AI Dispatcher Latency:** Fast 1.1s response using `gemini-3.5-flash-lite`.
- **Redis Chat Pipeline:** Sub-50ms message delivery with automated in-memory failover.
- **Security Posture:** 0 critical vulnerabilities identified by Snyk audit.

11. Team Contribution Table

Equitable Distribution of Engineering Work

Student ID	Member Name	Core Technical Roles & Contributions
0432320005101064	Md. Masud Rahman	Full-Stack System Architecture, Next.js Server Actions, Gemini AI Dispatcher, Upstash Redis Live Chat, SLA Calculation Engine.
0432320005101059	Rudro Antony Mrong	Requirements Engineering, SRS Formulation, System Modeling (DFD, Class, Sequence Diagrams), Literature Review & UI Design.
0432320005101088	Shafayat Hossain Shafin	System Tester & Quality Assurance, Automated Unit Testing with Vitest (19 test cases, 100% coverage), SLA Boundary Tests, Snyk Security Auditing, Test Documentation.

12. Current Project Limitations

Honest Academic Assessment of Boundary Constraints

Web-Only Access

- Currently operates as a responsive web application; lacks a native mobile app (Android/iOS) with background push notifications.

Manual Issue Reporting

- Relies on students and staff to manually report faults; no direct integration with automated hardware IoT sensors.

Cloud Infrastructure Dependence

- Requires active internet connectivity to communicate with Supabase, Upstash, and Google Gemini API endpoints.

Single-Campus Boundary

- Configured for single-campus university structure (UITS); multi-campus tenancy requires additional tenant isolation schemas.

13. Future Roadmap & Enhancements

Scalable Pathways for CampusCare 2.0

Native Mobile App

- Develop cross-platform React Native app with native push notifications, offline draft caching, and barcode scanning for lab equipment.

IoT Sensor Automation

- Integrate ESP32 / LoRaWAN sensors in restrooms and server rooms to automatically generate tickets for water leaks and overheating.

Predictive Analytics

- Machine learning models analyzing historical repair logs to forecast equipment failure before breakdowns occur.

14. Conclusion

Summary of Project Achievements

Summary of Deliverables

- Successfully engineered and deployed **CampusCare**, an automated, role-based university facility maintenance platform.
- Solved the historic challenges of scattered reporting, zero visibility, and lack of accountability across university operations.
- Delivered enterprise-grade features: **PostgreSQL RLS security**, **automated SLA overdue detection**, **Gemini-powered AI Dispatcher**, and **real-time chat with human takeover**.
- Fully validated through a 100% passing automated test suite and deployed on cloud infrastructure.

Final Verdict

CampusCare proves that modern web technologies and lightweight AI can transform campus operations into transparent, efficient, and student-centered environments.

Thank You!

We Welcome Questions & Feedback

Now Proceeding to Live System Demonstration

CampusCare: Intelligent University Facility Management Platform
Department of Computer Science & Engineering, UITS
Codebase & Documentation: `github.com/Shoytanbaba99/CampusCare`

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